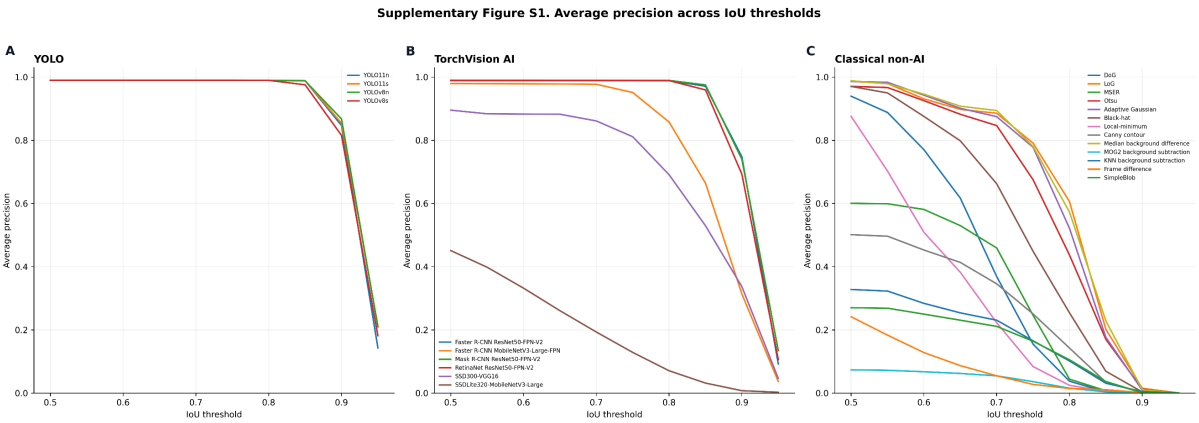
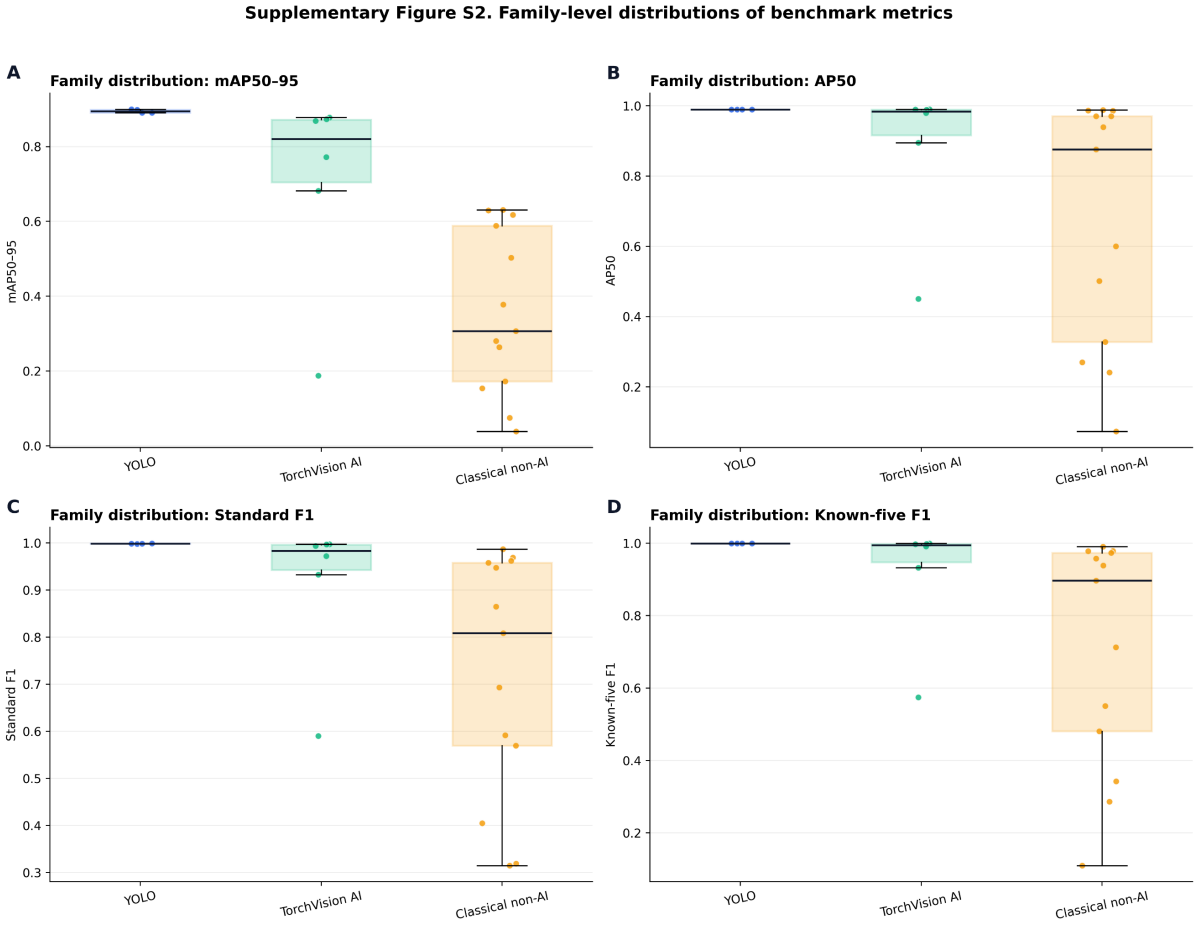
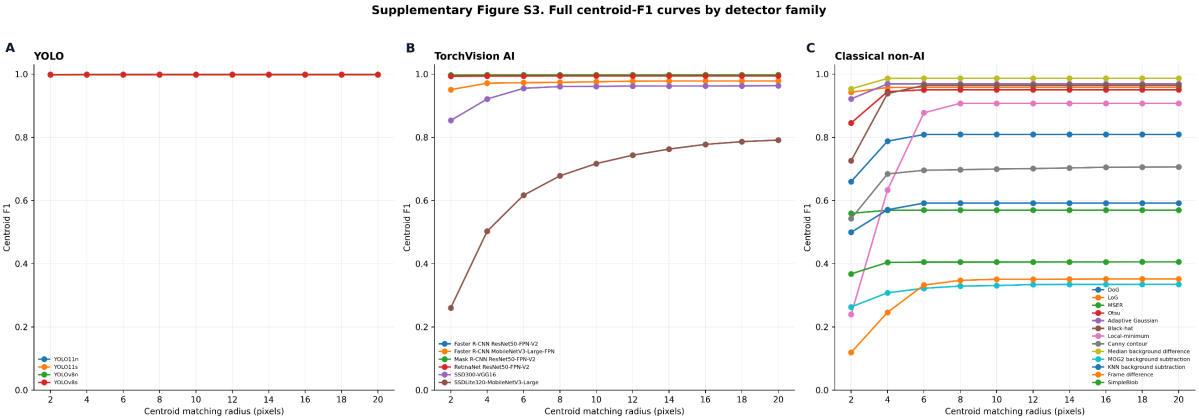


Supplementary Figure S1. Average precision across IoU thresholds from 0.50 to 0.95. (A) YOLO models. (B) TorchVision deep-learning models. (C) Classical methods.



Supplementary Figure S2. Family-level distributions of benchmark metrics. (A) mAP50–95. (B) AP50. (C) standard validation-selected F1. (D) known-five F1.



Supplementary Figure S3. Full centroid-F1 curves across matching radii from 2 to 20 pixels in 2-pixel increments. (A) YOLO models. (B) TorchVision deep-learning models. (C) Classical methods.